


ps 5.

$$X_i \sim \mathcal{N}(\mu, \sigma^2)$$

$$\mu \sim \mathcal{N}(\theta, \tau^2)$$

$$\mathcal{L}(\mu; x) = \left(\frac{1}{\sqrt{2\pi}\sigma^2} \right)^n \exp \left(-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 \right)$$

$$\pi(\mu) = \frac{1}{\sqrt{2\pi}\tau^2} \exp \left(-\frac{1}{2\tau^2} (\mu - \theta)^2 \right)$$

$$\begin{aligned} \pi(\mu; x) &\propto \exp \left(-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 - \frac{1}{2\tau^2} (\mu - \theta)^2 \right) \\ &= \exp \left(-\frac{1}{2\sigma^2} \left(\sum (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2 \right) - \frac{1}{2\tau^2} (\mu - \theta)^2 \right) \end{aligned}$$

$$\propto \exp \left(-\frac{n}{2\sigma^2} (\bar{x} - \mu)^2 - \frac{1}{2\tau^2} (\mu - \theta)^2 \right)$$

$$\propto \exp \left(-\left(\frac{n}{2\sigma^2} + \frac{1}{2\tau^2} \right) \mu^2 + \left(\frac{n}{\sigma^2} \bar{x} + \frac{1}{\tau^2} \theta \right) \mu + C \right)$$

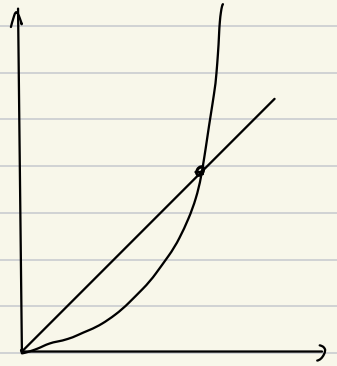
$$= \exp \left(-\frac{n}{2\sigma^2} - \frac{1}{2\tau^2} \right) \left(\mu - \frac{\frac{n}{\sigma^2} \bar{x} + \frac{1}{\tau^2} \theta}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}} \right)^2$$

$$= \exp \left(-\frac{n\tau^2 + \sigma^2}{2\sigma^2\tau^2} \right) \left(\mu - \frac{\theta\sigma^2 + n\tau^2\bar{x}}{n\tau^2 + \sigma^2} \right)$$

$$\mu|x \sim \mathcal{N} \left(\frac{\sigma^2 + n\tau^2}{\theta\sigma^2 + n\tau^2\bar{x}}, \frac{\tau^2\sigma^2}{\sigma^2 + n\tau^2} \right)$$

Past Paper.

$$\begin{aligned} 1 \quad (a) \quad P(Y < 3X) &= \int_0^3 \int_{x^2}^{3x} \frac{1}{x} e^{-\frac{y}{x}} dy dx \\ &= \int_0^3 \left. \frac{1}{x} \cdot (-x) \cdot e^{-\frac{y}{x}} \right|_{x^2}^{3x} dx \\ &= \int_0^3 (-e^{-3} + e^{-x}) dx \\ &= -3e^{-3} + 1 - e^{-1} \\ &= 1 - 4e^{-3} \end{aligned}$$



$$\begin{aligned} (b) \quad f_x(x) &= \int_{x^2}^{+\infty} \frac{1}{x} \exp\left(-\frac{y}{x}\right) dy \\ &= \left. \frac{1}{x} \cdot (-x) \exp\left(-\frac{y}{x}\right) \right|_{x^2}^{+\infty} \\ &= -1 \cdot (-e^{-x}) = e^{-x} \end{aligned}$$

$$(c) \quad f_{Y|X}(y|x) = \frac{f_{X,Y}}{f_X} = \frac{1}{x} \exp\left(-\frac{y}{x} + x\right)$$

$$\begin{aligned} P(Y > 6 | X=2) &= \int_6^{+\infty} \frac{1}{2} \exp\left(-\frac{y}{2} + 2\right) dy \\ &= \left. \frac{1}{2} e^2 \cdot (-2 e^{-\frac{y}{2}}) \right|_6^{+\infty} \\ &= \frac{1}{2} e^2 \cdot 2 \cdot e^{-3} \\ &= \frac{1}{e} \end{aligned}$$

$$(d) \quad E[Y|X] = \int_{x^2}^{+\infty} y \cdot \frac{1}{x} \exp\left(-\frac{y}{x} + x\right) dy$$

$$\begin{aligned}
&= \frac{e^x}{x} \int_{x^1}^{+\infty} y e^{-\frac{y}{x}} dy \\
&= \frac{e^x}{x} \cdot e^{-\frac{y}{x}} (xy - x^2) \Big|_{x^1}^{+\infty} \\
&= \frac{e^x}{x} e^{-x} (x^2 + x^1) \\
&= x^2 + x
\end{aligned}$$

$$\begin{aligned}
(e) \quad E[Y] &= E[E[Y|X]] = \int_0^{+\infty} (x^2 + x) e^{-x} dx \\
&= 2! - 1! = 3
\end{aligned}$$

$$(f) \quad P(Y \geq 8) \leq \frac{E[Y]}{8} = \frac{3}{8}$$

$$\begin{aligned}
2. (a) \quad E[X] &= \int_3^{+\infty} x \cdot \frac{f_1}{x^4} dx & E[X^2] &= \int_3^{+\infty} x^2 \cdot \frac{f_1}{x^4} dx & \sigma^2 &= 27 - \frac{81}{4} \\
&= 81 \left(-\frac{1}{2} \frac{1}{x^2} \right)_3^{+\infty} & &= \int_3^{+\infty} \frac{f_1}{x^2} dx & &= \frac{27}{4} \\
&= \frac{81}{2 \cdot 9} = \frac{9}{2} & &= 81 \left(-\frac{1}{x} \right)_3^{+\infty} = 27
\end{aligned}$$

$$\begin{aligned}
P(X \geq 12) &= P\left(|X - \frac{9}{2}| > \frac{15}{2}\right) \leq \frac{27}{4} \left(\frac{2}{15}\right)^2 \\
&= \frac{27^3}{15 \cdot 25} = \frac{3}{25}
\end{aligned}$$

$$\begin{aligned}
(c) \quad M_Y(t) &= E[e^{tY}] = \int_0^{+\infty} e^{ty} \cdot \frac{1}{\sqrt{2\pi}y^3} \exp\left(-\frac{(y-\phi)^2}{2\phi^2 y}\right) dy \\
&= \int_0^{+\infty} \frac{1}{\sqrt{2\pi}y^3} \exp\left(ty - \frac{1}{2\phi^2}y + \frac{1}{\phi} - \frac{1}{2y}\right) dy
\end{aligned}$$

$$(d) E[Y] = M_Y'(0) \quad M_Y'(t) = \frac{1}{\phi} \left(-\frac{-2\phi^2}{\sqrt{1-2\phi^2 t}} \cdot \frac{1}{2} \right) \exp\left(\frac{1}{\phi}(1+\sqrt{1-2\phi^2 t})\right)$$

$$E[Y^2] = M_Y''(0)$$

$$\begin{aligned} 3. (a) E[X] &= \int_0^1 x [(1-\theta) + 2\theta x] dx \\ &= \left[\frac{1}{2} (1-\theta)x^2 + \frac{2}{3} \theta x^3 \right]_0^1 \\ &= \frac{1}{2} (1-\theta) + \frac{2}{3} \theta \\ &= \frac{1}{2} + \frac{1}{6} \theta = \bar{X} \quad \theta = 6\bar{X} - 3 \end{aligned}$$

$$(b) MSE = \text{Var}(\hat{\theta}) + \beta_{bias}^2$$

$$E[X] = \frac{1}{3} + \frac{\theta}{6}$$

$$\text{Var}(X) = \frac{1}{12} - \frac{\theta^2}{36}$$

$$\text{Var}(\bar{X}) = \frac{1}{n} \left(\frac{1}{12} - \frac{\theta^2}{36} \right) = MSE(\hat{\theta})$$

$$(c) F(x) = \int_0^{\infty} \left(\frac{1}{2} + t \right) dt = \frac{1}{2} (x^2 + x)$$

$$F(k_2) = \frac{1}{2} \quad F(k_1) = 1 - \alpha$$

$$(d) P(X \leq k | H_0) = k = \alpha$$

$$P(X \leq k) = \int_0^{\alpha} ((1-\theta) + 2\theta x) dx = \alpha + \theta(\alpha^2 - \alpha)$$

4 (a)	1	2	3	4	5	6	7	8	9	10
	-0.3	0.2	-2.6	0.1	-1.8	0.6	-1.5	-2.0	-2.3	-12.4
	3	2	9	1	6	4	5	7	8	10

$$w^+ = 1 + 2 + 4 = 7$$

$$w^- = 3 + 9 + 6 + 5 + 8 + 10 = ?$$

$$\min(w^+, w^-) = 7$$

some evidence.

rb)

$$5. \quad 1a) \quad L(\theta|x) = \binom{n}{x_1} (1-\theta)^{x_1} \binom{n-x_1}{x_2} \left(\frac{\theta}{2}\right)^{x_2} \cdot \left(\frac{\theta}{2}\right)^{x_3}$$

$$\begin{aligned} \pi(\theta|x) &\propto (1-\theta)^{x_1} \left(\frac{\theta}{2}\right)^{x_2} \left(\frac{\theta}{2}\right)^{x_3} \theta^{\alpha-1} (1-\theta)^{b-1} \\ &\propto (1-\theta)^{x_1+b-1} \theta^{x_2+x_3+\alpha-1} \end{aligned}$$

$$\pi(\theta|x) \propto \beta(\alpha + x_1 + x_3, b + x_1)$$

$$1b) \quad \hat{\theta} = E(\theta|x) = \frac{\alpha}{\alpha+\beta} = \frac{2}{3}$$

$$1c) \quad \beta(4, 2) = \frac{\Gamma(4)\Gamma(2)}{\Gamma(6)} = \frac{1}{20}$$

$$M_X(t) = E[e^{tx}]$$

$$= E\left[\sum_{n=0}^{\infty} \frac{(tx)^n}{n!}\right]$$

$$M^{(n)}(t) = E\left[X^n \sum_{n=0}^{\infty} \frac{(tx)^n}{n!}\right]$$

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1)$$

$$\sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x)$$